

ASTRAM Congestion Copilot

Forecasting event-driven congestion — and the plan to clear it

Gridlock Hackathon 2.0 · Round 2 (Prototype) · Theme 2

Event-Driven Congestion (Planned & Unplanned)

THE PROBLEM



Events break traffic faster than crews can react

- ▶ Rallies, festivals, sports, construction & sudden gatherings cause localized breakdowns.
- ▶ Today: event impact isn't quantified in advance.
- ▶ Resource deployment is experience-driven, not data-driven.
- ▶ No post-event learning system to improve next time.

Theme 2 asks two things — forecast impact AND recommend manpower, barricading & diversion. Most stop at prediction. We do both.

OUR SOLUTION



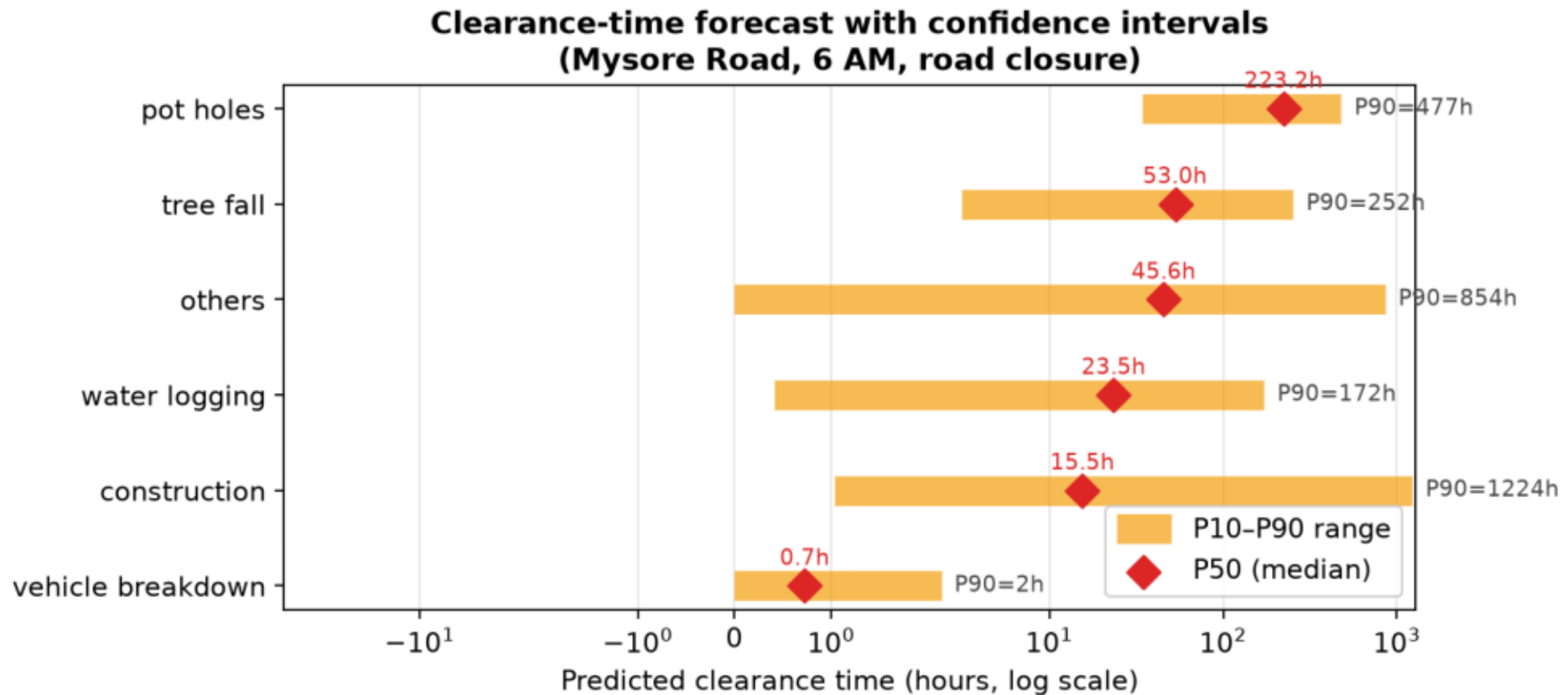
A working forecast-to-dispatch prototype

- ▶ FORECAST — clearance time as a P10 / P50 / P90 interval, not a fragile guess.
- ▶ RECOMMEND — response team, crew count, equipment, barricading & diversion.
- ▶ DISPATCH — shift-level pre-positioning plan by corridor × time window (CSV/JSON).
- ▶ PREVENT — recurring waterlogging / pothole / construction hotspots.

Live Streamlit app · trained on real ASTRAM / BTP event data.

FORECAST

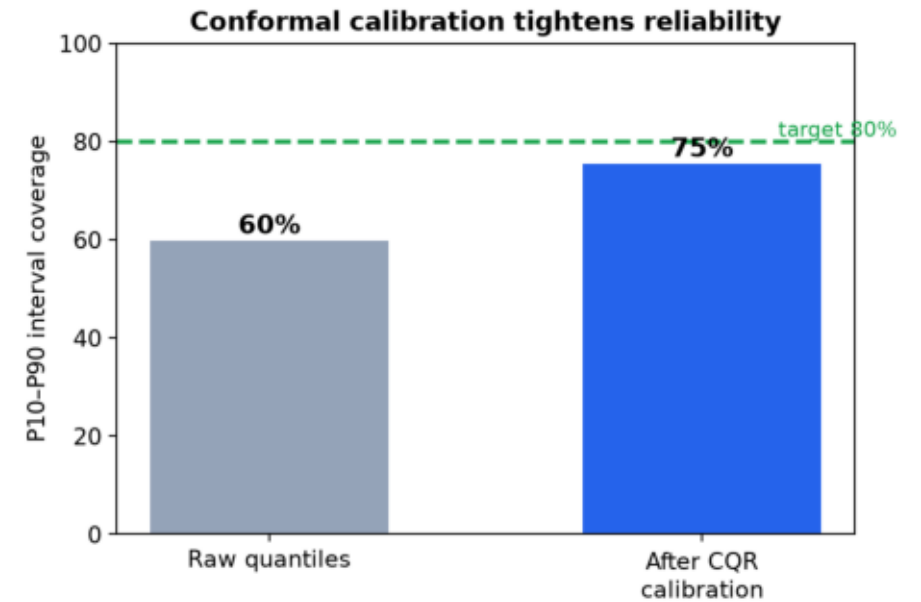
Why a range beats a single number



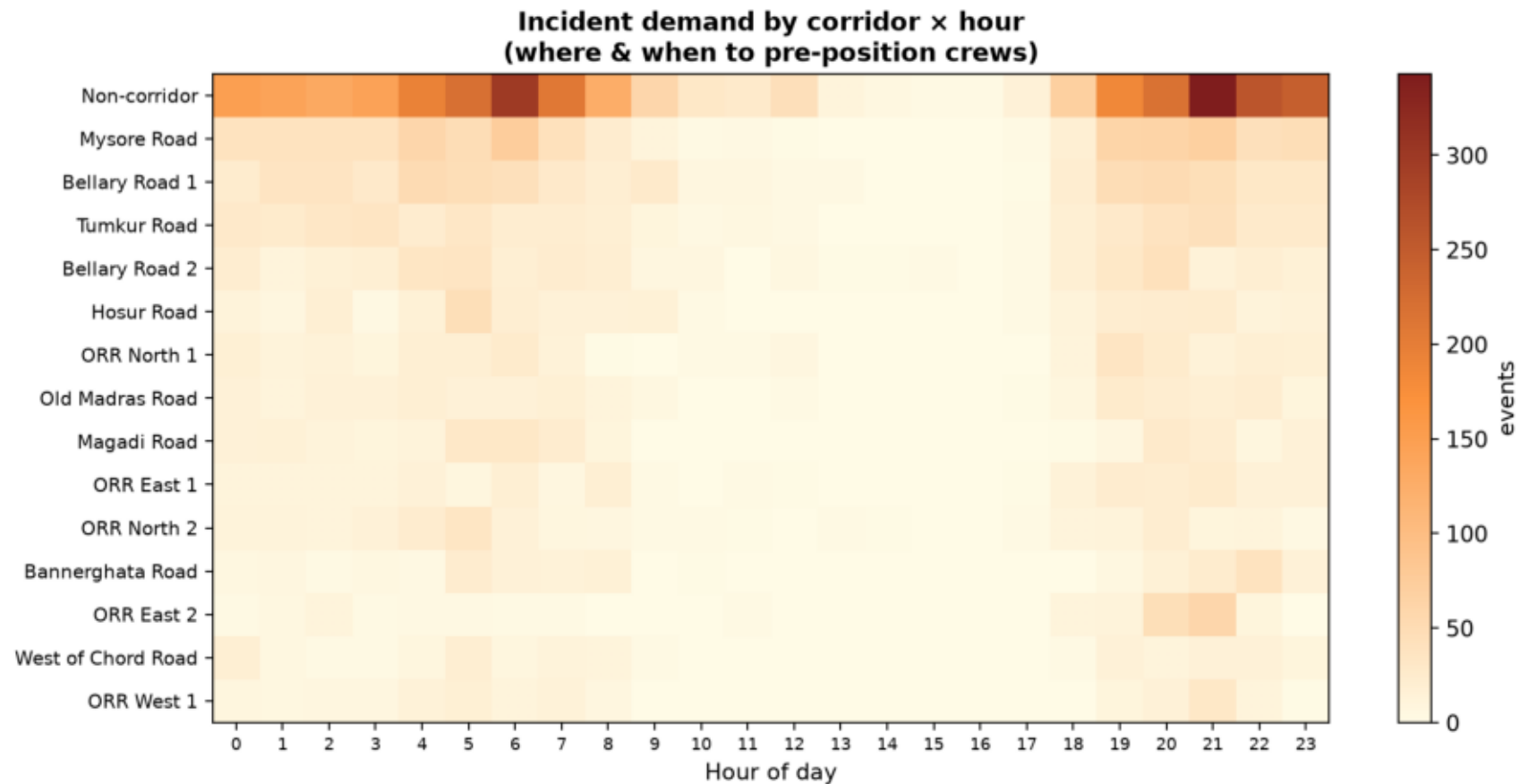
THE MODEL

Calibrated, leakage-safe, honest

- ▶ LightGBM on log-resolution + quantile models (P10/P50/P90).
- ▶ Dedicated models for high-variance causes (potholes, construction).
- ▶ Conformalized Quantile Regression calibrates the intervals.
- ▶ Strict temporal validation — train on earlier events, test on later.



Where & when to pre-position crews



Honest numbers (temporal hold-out)

1.4 h

Typical (median) error

~0.7 h

Vehicle breakdown P50
(60% of all events)

75%

P10-P90 coverage
(60% → after CQR)

Mean error looks large only because rare multi-day infrastructure incidents dominate the average — which is exactly why we report intervals. Surfaced in the app's Model Card, not hidden.

From incident to action in one screen

- ▶ Breakdown, Mysore Rd, 6 AM → ~0.7 h, HIGH, 2 towing units.
- ▶ Waterlogging + closure → ~24 h (P90 ~170 h), CRITICAL, pumps + diversion.
- ▶ Dispatch tab → demand heatmap + downloadable staging plan.
- ▶ Hotspots tab → preventive-maintenance targets.

Built on real BTP data · deployable today · scales to live ASTRAM feeds.

Thank you.